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AI Lab # 6

This lab explores optimization-based search strategies by implementing Hill Climbing and Genetic Algorithms to iteratively improve solutions in state-space problems.

Artificial Intelligence (LAB)

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Today's Agenda

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01 What is Hill Climbing?

02 Algorithm Steps and Example

03 What is Genetic Algorithms?

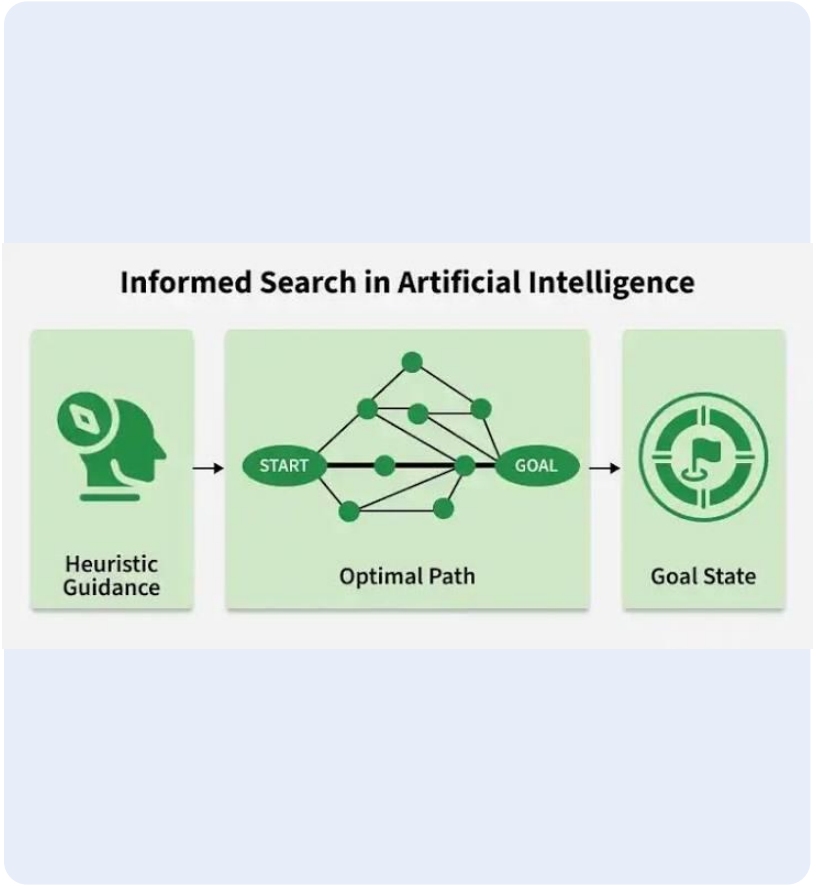
04 Algorithm Steps and Example

05 Lab Tasks

Informed Search

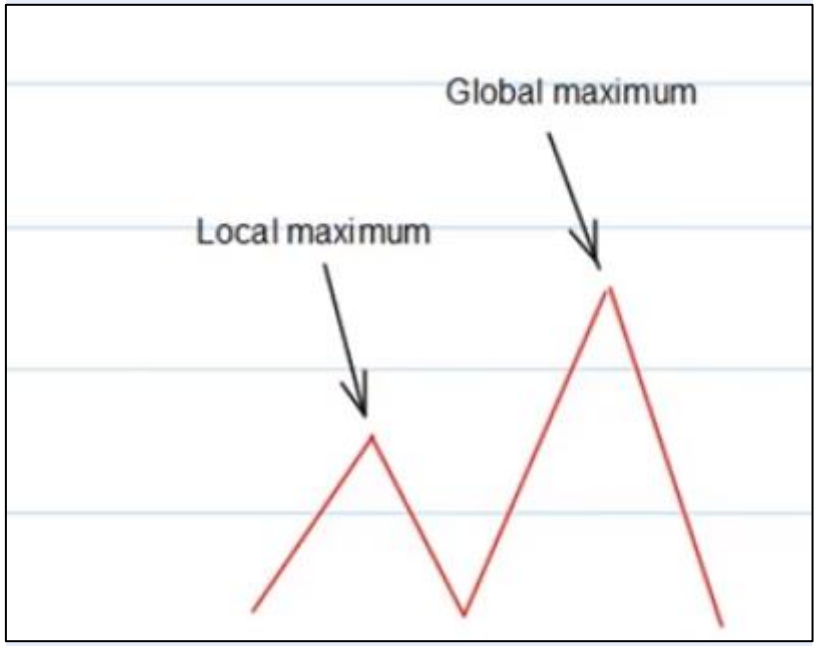
- Informed search (also called ***heuristic search***) uses **extra information** (called a ***heuristic***) to find solutions faster and smarter than uninformed methods like BFS, DFS, or UCS.
- A heuristic is like a guess or clue that tells the algorithm how close you are to the goal.

Imagine you're playing hide and seek, and someone tells you: "I'm here".



What is Hill Climbing?

- A simple optimization algorithm that starts with a random solution and then repeatedly moves to a “better” solution.
- Finds quick solution, works well for certain problems like TSP, traveling salesman problem.
- Finds local optimal solution not global optimal. Example: Finds ***“good” answer from the paper***, but not the ***“best” answer from the book***.

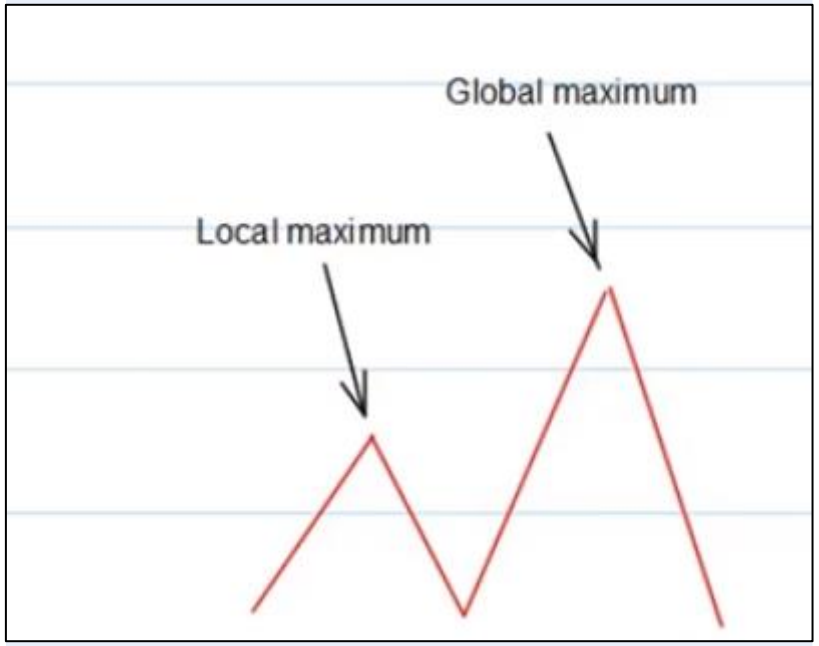


Imagine climbing a hill in fog:

- You cannot see the whole mountain.
- You only feel which direction is going upward.
- You take steps toward higher ground.
- You stop when every direction seems lower.

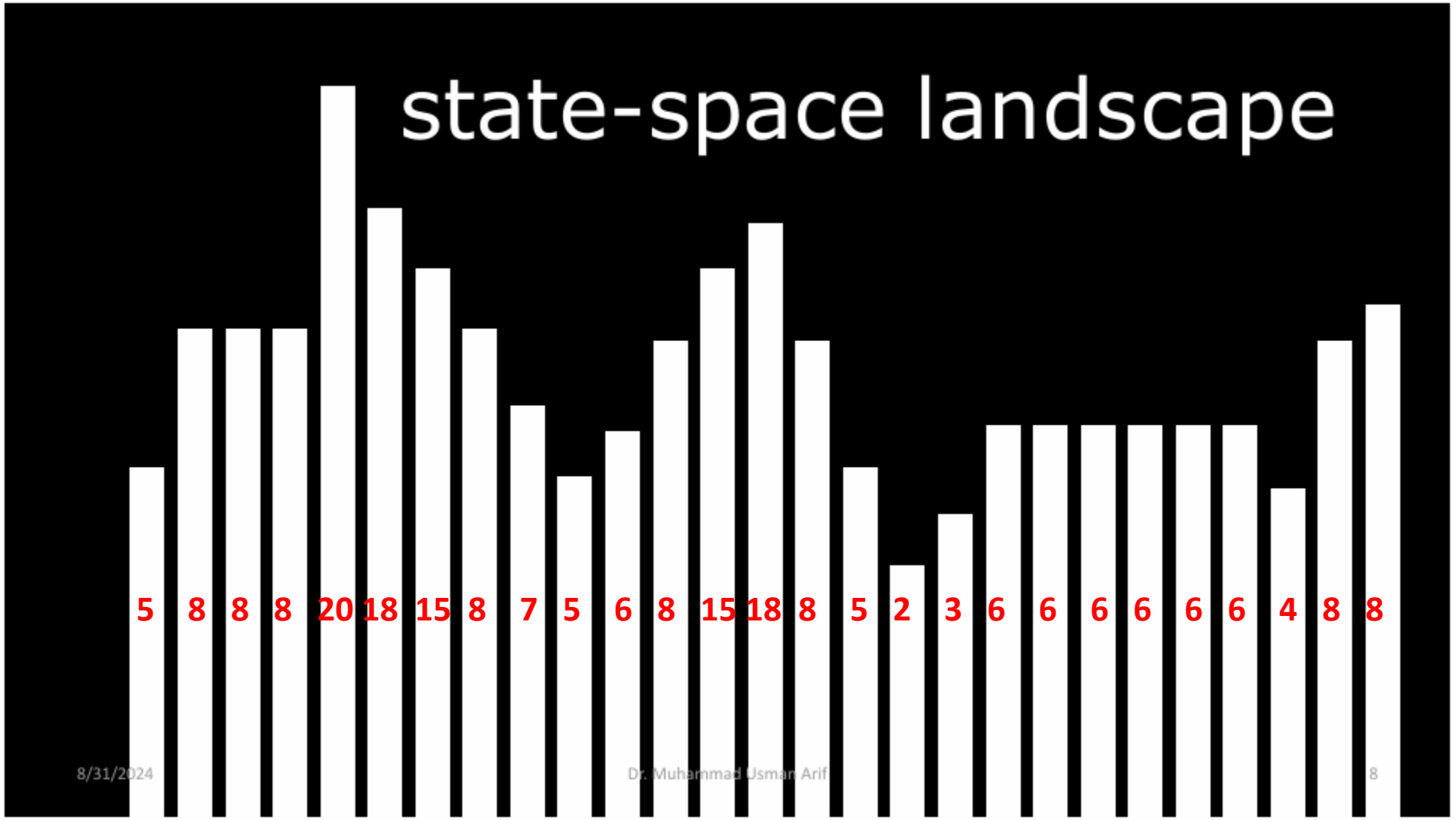
Applications of Hill Climbing

- GPS Navigation – to quickly reach a destination using shortest-looking turns.
- Robot pathfinding – to move toward a goal using nearest steps.
- Internet routing – selecting fastest-looking paths for data packets.
- Job scheduling – picking the task that looks quickest to finish.
- Machine Learning (ML) Optimization – Used for hyperparameter tuning to maximize the model's accuracy score.
- Traveling Salesman Problem (TSP) – Finding a near-optimal, short route by locally swapping the order of cities.
- Game AI – Evaluating and choosing the locally best next move in complex board games (e.g., Chess) to achieve a high heuristic score.
- Neural Network Weight Adjustment – Fine-tuning network weights by making small changes to minimize the training error function



Algorithm Steps

1. **Start with an initial solution:** Begin at a random or given starting node (current state)
2. **Evaluate neighbors:** Look at all neighboring nodes and calculate their heuristic value h(n)
3. **Pick the best neighbor:** Select the neighbor with the best heuristic value
4. **Compare with current node:** If the best neighbor is better than the current node, move to it. If not, stop — you've reached a peak (local or global optimum)
5. **Repeat until no improvement is possible:** Keep repeating steps 2 to 4 until every neighbor is worse than (or equal to) the current node, then return the final solution


state-space landscape
5 8 8 8 20 18 15 8 7 5 6 8 15 18 8 5 2 3 6 6 6 6 6 6 4 8 8

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Example of Hill Climbing

Now that we've covered Hill Climbing, let’s move towards practical code-based example

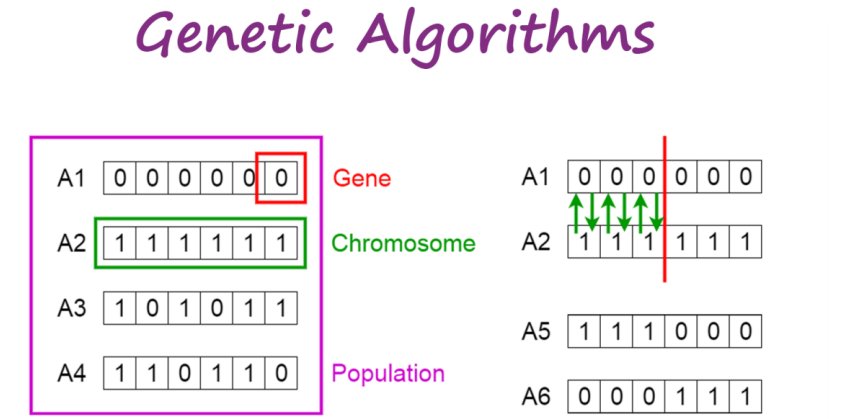
Hill Climbing

Hill Climbing on nodes A, B, C, D, E, F, G, H, I, J, K, L, M, N, O. Starting node is S and Goal node is M.

```
graph = { 'S': ['A', 'B'], 'A': ['C', 'D', 'E'], 'B': ['F', 'G'], 'C': ['H', 'I'], 'G': ['J', 'K'], 'I': ['L', 'M'], 'K': ['N', 'O’] }
heuristic = { 'S': 12, 'A': 9, 'B': 11, 'C': 8, 'D': 9, 'E': 7, 'F': 9, 'G': 9, 'H': 6, 'I': 5, 'J': 7, 'K': 6, 'L': 2, 'M': 0, 'N': 4, 'O’: 4 }

def hill_climbing(start):
    current = start
    path = [current]
    while True:
        neighbors = graph.get(current, [])
        if not neighbors:
            break

        best = neighbors[0]
        best_value = heuristic[best]
        for n in neighbors:
            if heuristic[n] < best_value:
                best_value = heuristic[n]
                best = n
        if heuristic[best] >= heuristic[current]:
            break
        current = best
        path.append(current)

    print("Hill-Climbing Path without back tracking:", path)

hill_climbing('S')
```

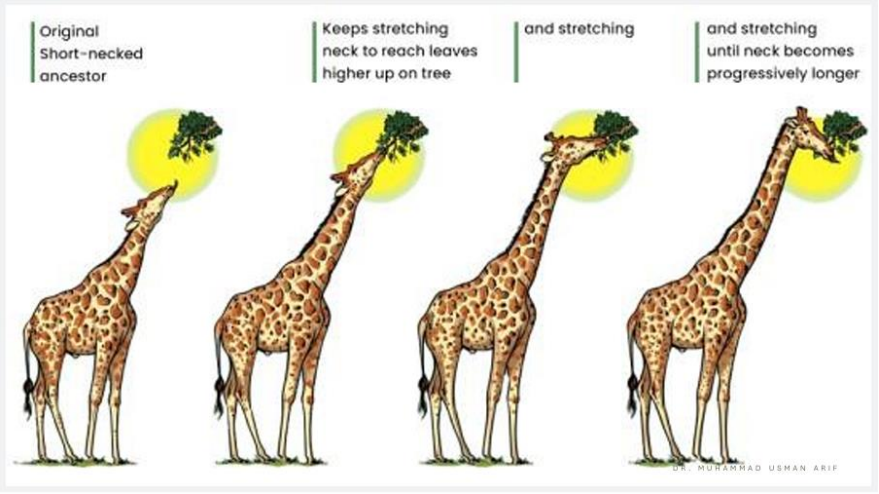
What is Genetic Algorithms?

- Concept: “Can computers solve problems the same way nature solves them?”
- Genetic Algorithms borrow the language of biological evolution: genes, chromosomes, survival of the fittest, and generations.
- Over many generations, solutions evolve toward the best one.



Core Concepts:

- **Population:** Start with possible solutions, like creatures in nature.
- **DNA:** Each solution carries parameters that describe it.
- **Crossover:** The best solution survives and combine.
- **Mutation:** A random change explores a new possibility.



Algorithm Steps

1. **Initialize population:** Start with a set of random solutions (e.g. 4 random numbers between 0 and 31), each treated as an individual's "DNA"

2. **Calculate fitness and Selection:** Evaluate each individual using a fitness function to measure how good the solution is (e.g., fitness = the number itself, bigger is better). Sort the population and pick the top individuals (parents) with the best fitness values

3. **Crossover:** Combine parts of the two parents' binary representations at a random or chosen crossover point to produce a new child

4. **Mutation:** With a small probability (e.g., 30%), randomly flip a bit in the child to introduce variation and explore new possibilities

5. **Replacement:** Replace the worst individual in the population with the new child

6. **Repeat until convergence:** Repeat steps 2 to 6 for several generations until the optimal solution is found or a fixed number of generations is reached

Example Problem: We want to find the largest number between 0 and 31

- **Population**: Start with 4 random numbers. **[5, 12, 7, 20]**
- **Fitness:** The bigger the number → the better.
- **Selection**: Pick the top 2 numbers. **20 and 12**
- **Crossover:** Mix digits (binary) of the two numbers to create a new number.
  - **20 = 10 | 100**
  - **12 = 01 | 100**
  - **20: 10100 → child**
- **Mutation:** Randomly change one bit to explore new numbers. **10100 → 10101 (21)**
- **Replace worst number** in population with new child. Take the new number (child) and replace the smallest number in the list.
  [5, 12, 7, 20] → child = 21 → replace 5 → [21, 12, 7, 20]

Tips

- In Python the **| is a bitwise OR operator,** but when we use it like this in crossover, its effect **acts like "combining bits"**.
- **There is no strict rule:** the **crossover** point can be **fixed or random**.

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Example of Genetic Algorithms

Now that we've covered Genetic Algorithms, let’s move towards practical code-based example

Genetic Algorithms

Genetic Algorithms for random population for size of 32.

```
import random

population = [random.randint(0, 31) for _ in range(4)]
print("Initial population:", population)

for generation in range(10):
    population.sort(reverse=True)  # bigger = better
    parent1, parent2 = population[0], population[1]
    child = (parent1 & 0b11100) | (parent2 & 0b00011)  # simple mix
    if random.random() < 0.3:
        child ^= 1
    population[-1] = child
    print(f"Gen {generation+1}: {population}")
    if 31 in population:
        print("Solution found!")
        break
```

ACTIVITY

Lab Task: Hill Climbing & Genetic Algorithms (Informed Search)

- *Implement the Hill Climbing algorithm to solve the 8-queens problem and display the number of attacking pairs at each move until a local optimum is reached.*
- *Write a simple Genetic Algorithm that tries to maximize a fitness function of your choice (such as number of 1s in a 5-bit binary string). Show population updates for at least 5 generations.*

Thank You

Questions or discussions are welcomed!